"On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this ekam. The work done on this ekam is totally my own. I understand that by the school code, violation of these principles will leas to a zero grade and is subject to harsh discipline issues."

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1) Consider the matrix $A = \begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix}$. Some of the elimination matrices are given below:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & k & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ -8 & 1 & 0 & 0 \\ m & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1/2 & 1 \end{bmatrix}$$

a) (5pts) Find the unknow entries $\alpha, \gamma, \beta, k$ and $m = ?$

b) (5pts) find the eliminated upper triangle matrix $U = ?$

c) (10 pts) Determine the remaining elimination matrices(s).

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answers

(a)
$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ -8 & 1 & 0 & 0 \\ m & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix} \begin{matrix} R_3 - R_2 \to R_3 \\ \longrightarrow \\ R_4 - \frac{1}{8} R_2 \to R_4 \\ R_2 + 8R_1 \to R_2 \end{matrix}$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & k & 0 & 1 \end{pmatrix} \Rightarrow m + 8 = 0 \; ; \quad \boxed{m = -8}$$

$$\boxed{k = -1/8}$$

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2) A geometry in 'k' dimensional space is given by $x=5y$ expression, where 'x' and 'y' represents the axis. The intersection with another geometry gives a volume

a) Give the minimum dimension of the space with your valuable comments.

b) Propose an expression for the unknown geometry to have a volume in intersection

c) I proposed a geometry $x+4y-z=6$ find all vectors in the intersection. Do these vectors form a vector space? The intersection has how many dimensions inside how many dimensional space

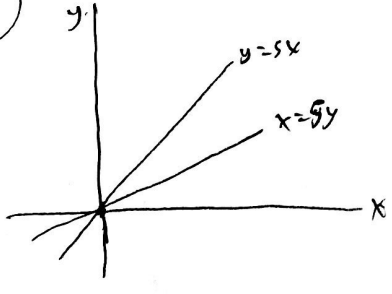
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answers

(a) $x=5y$ is $(k-1)$ dimensional in $k$-space. The intersection of two such geometry is $(k-2)$-dimensional in order to have a volume $k-2$ should be at least 3D. $\Rightarrow$ $k-2 \geq 3$

$$\boxed{k \geq 5}$$

(b)




$\rightarrow z$ when $k=2$, the intersection is a point (zero dimension)

when $k=5$, $y=5x$ and $x=5y$ form a volume

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2 (c)

$$\left.\begin{array}{l} x+4y-z=b \\ x=5y \end{array}\right\} \Rightarrow$$

$5y+4y-z=b \;/\; 9y-z=b$

$\Rightarrow z=9y-b$

Since $k=3$ / let $u\,v$ be after axes

$$\begin{pmatrix} x \\ y \\ z \\ u \\ v \end{pmatrix} = \begin{pmatrix} 5y \\ y \\ 9y-b \\ u \\ v \end{pmatrix} = y\begin{pmatrix} 5 \\ 1 \\ 9 \\ 0 \\ 0 \end{pmatrix} + u\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} +$$

$$v\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ -b \\ 0 \\ 0 \end{pmatrix} \Leftarrow \text{intersection vectors}$$

The intersection has 3 dimensions inside 5-dimensional space

They form a vector space in $\mathbb{R}^3$

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3) Suppose $u$ is a unit vector in $R^n$. This problem is about the $n$ by $n$ symmetric matrix $H = I - 2uu^T$:

a) Show directly that $H^2 = I$

b) Show that $H$ is an orthonormal matrix

c) One eigenvector of $H$ is $u$ itself. Find the corresponding eigenvalue.

d) Is $H$ diagonalizable? Why or why not?

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Answers

(a) $H = I - 2uu^T$

$H^2 = I^2 - 4Iuu^T + 4(uu^T)^2$

$= I - 4uu^T + 4(uu^T)^2$

$uu^T$ is a projection matrix, so

$(uu^T)^2 = uu^T$

$= I - 4uu^T + 4uu^T$

$= I$

(b) if $H^TH = I$ and $\det(H) = \pm 1$, then $H$ is orthonormal

$\det(H^2) = \det(H) \times \det(H) = 1$

$\det(I) = 1$

$\Rightarrow (\det(H))^2 = 1$

$\boxed{\det(H) = \pm 1}$

$H^T = I^T - 2(uu^T)^T = I - 2(u^T)^T u^T = H$

So, $\boxed{H^TH = H^2 = I}$

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3 (c) $Hu = Iu - 2uu^T u$

$Hu = -u \rightarrow u$ is an eigenvector

which has the comes paneling eigenvalue of

$\boxed{\lambda = -1}$

3 (d) $I$ has $(n-1)$ eigenvalues 1 with eigenvectors which are linearly independent vectors orthogonal to $u$, since $Hx =$

$Ix - 2uu^T x = x$ for $x \perp u$

Therefore, $H$ cannot be diagonalized because it does not have $n$ distinct eigenvalues

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(7)

4) Two tanks are connected, first tank has $v(0) = 10$ liters liquid and the second tank has $w(0) = 40$ liters of water. The liquid diffusion between the tanks are proportional to the difference $v - w$.

a) Which one of the following differential equation set expresses the process?

$$\begin{bmatrix} v_{k+1} \\ w_{k+1} \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} v_k \\ w_k \end{bmatrix} \qquad \frac{d}{dt}\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix} \qquad \frac{d}{dt}\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix}$$

b) Determine percentages of the mixture in tank1 $t=1$ sec

c) Find how much litres of liquids are exist in each tank in steady state?

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Answers

(a) $\frac{dv}{dt} = -v + w = -(v - w)$

$\frac{dw}{dt} = v - w$

$$\frac{d}{dt}\begin{pmatrix} v \\ w \end{pmatrix} = \begin{pmatrix} w - v \\ v - w \end{pmatrix} = \underbrace{\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} v \\ w \end{pmatrix}}_{\text{Second one}}$$

(b) Eigen values of $\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$:

$$\begin{vmatrix} -1-\lambda & 1 \\ 1 & -1-\lambda \end{vmatrix} = (\lambda+1)^2 - 1 = 0$$

$$\boxed{\lambda_1 = 0, \ \lambda_2 = -2}$$

eigenvectors are $v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $v_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$,

so, $\begin{pmatrix} v \\ w \end{pmatrix} = c_1 e^{\lambda_1 t}\begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 e^{\lambda_2 t}\begin{pmatrix} -1 \\ 1 \end{pmatrix}$

$\begin{pmatrix} v \\ w \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 e^{-2t}\begin{pmatrix} -1 \\ 1 \end{pmatrix}$

at $t=0 \Rightarrow \begin{pmatrix} v \\ w \end{pmatrix} = \begin{pmatrix} 10 \\ 40 \end{pmatrix} = \begin{pmatrix} c_1 + c(-c_2) \\ c_1 + c_2 \end{pmatrix}$

$$\Rightarrow \boxed{c_1 = 25, \ c_2 = 15}$$

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(b) So,

$v(t) = 25 - 15e^{-2t}$

$w(t) = 25 + 15e^{-2t}$

$v(t=1) = 25 - 15e^{-2} \simeq 23$

$\frac{23}{40+10} = \boxed{46\%}$ at mixture in tank 1

(c) As $t \to \infty$, $e^{-2t}$ goes to zero

$\rightarrow v(t) = 25$ liters

$w(t) = 25$ liters

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